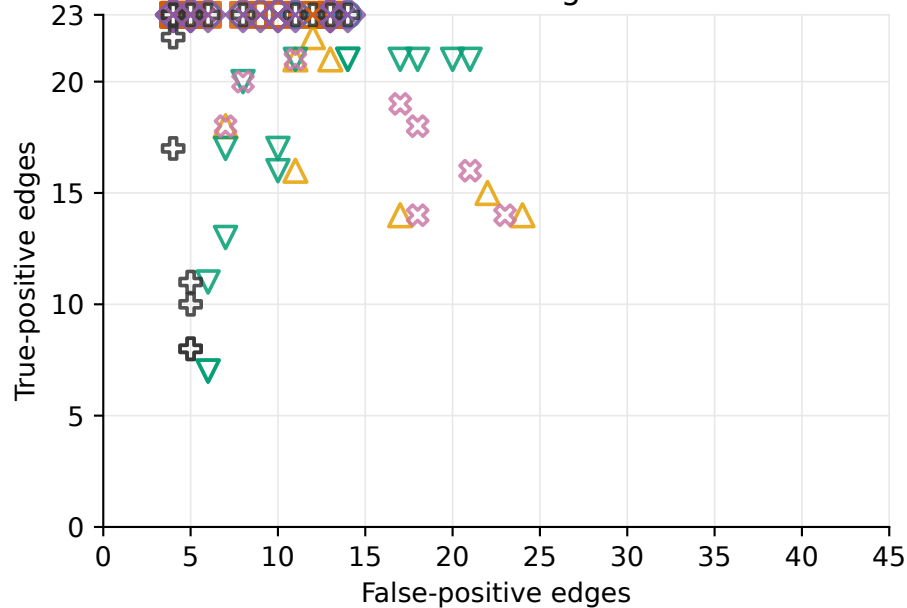


Causal Chambers: seven-condition pilot

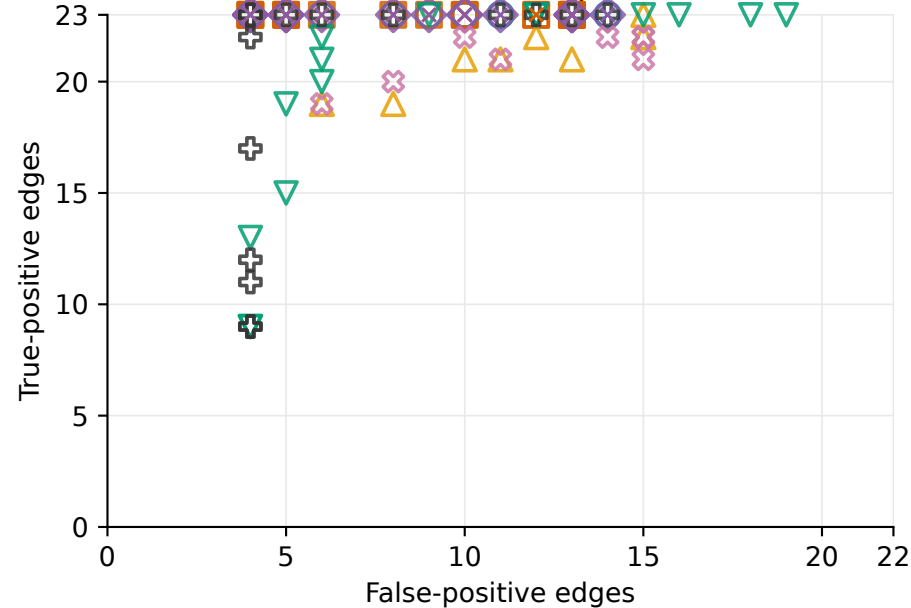
97/97 successful fits | 40,000 observations per configuration | no rank transformation

Actuator chain (scm_4)

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- PS-MIP: hidden targets (diagnostic)
- UT-IGSP: targets supplied

- GnIES: target union supplied
- GIES: complete targets
- IGSP: complete targets

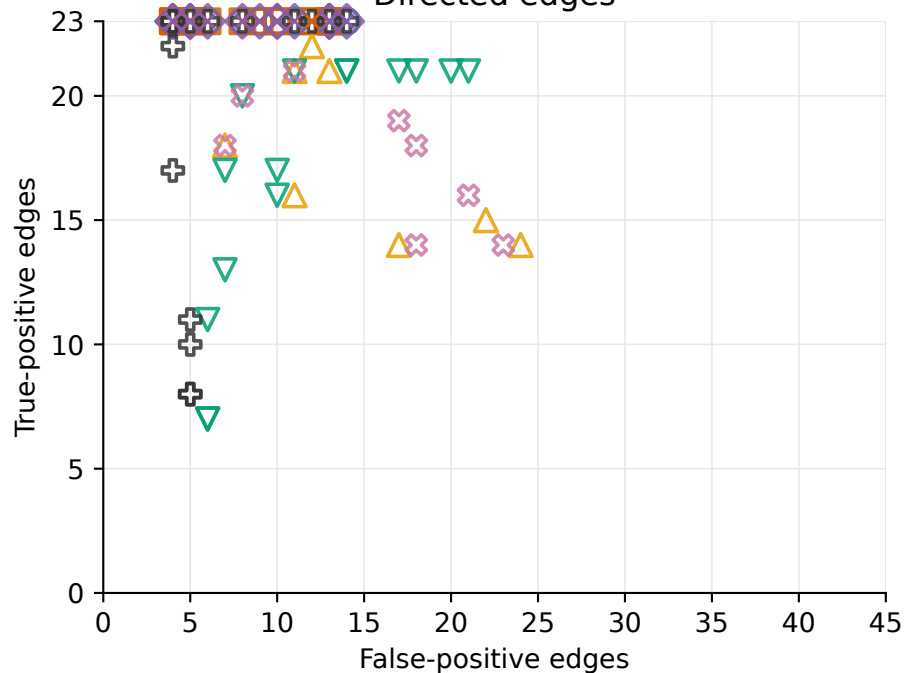
Reference: 21 physical + 2 programmed links. Full paths; no lines, envelope, or AUC.

Causal Chambers | Actuator chain (scm_4)

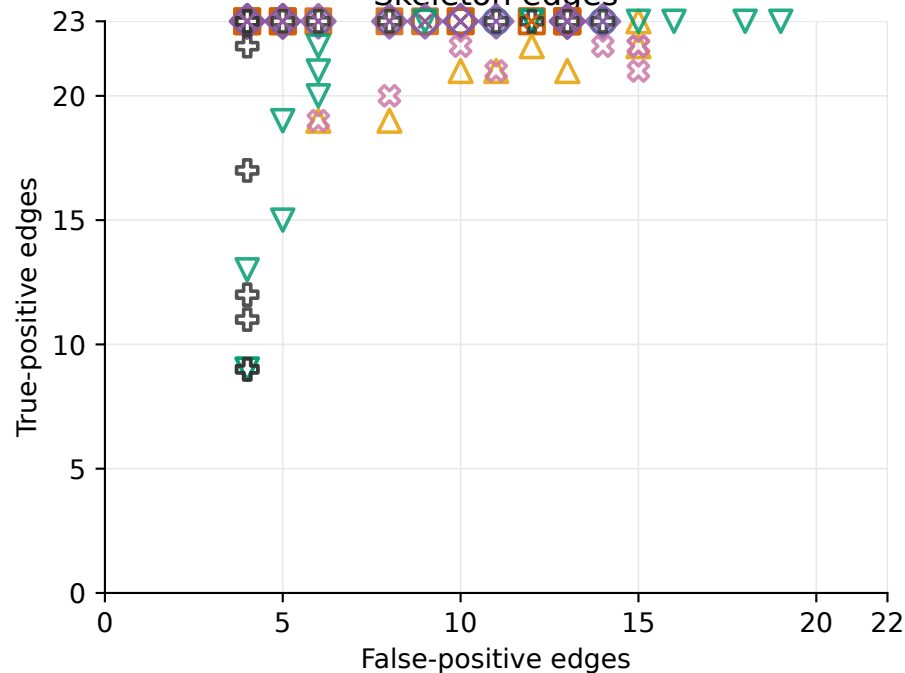
40,000 observations | 10 variables | 23 reference edges | 97/97 successful fits

Full-range paths; every successful operating point retained

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied

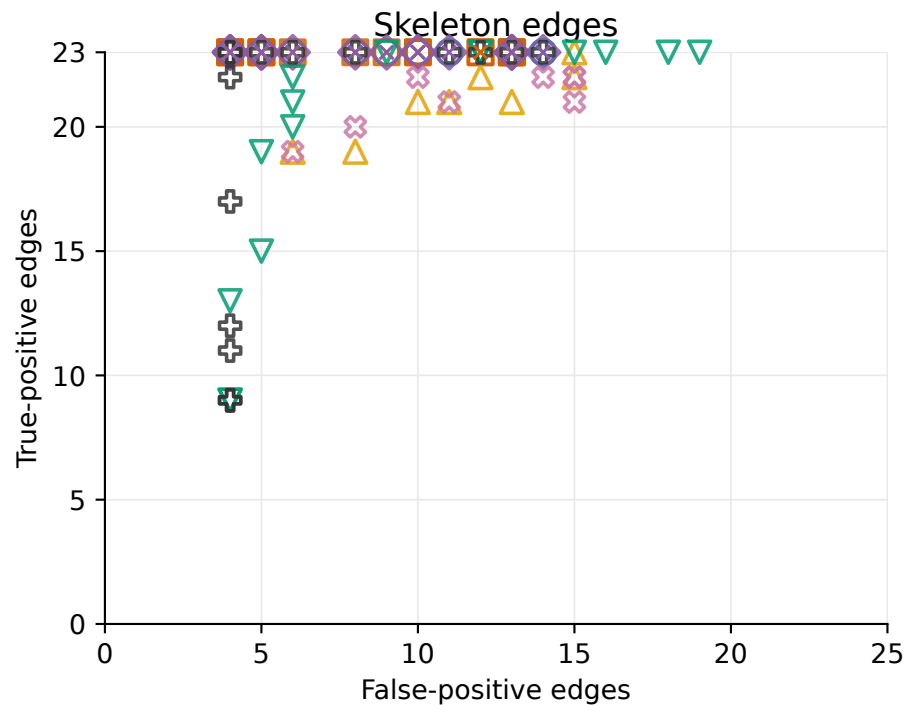
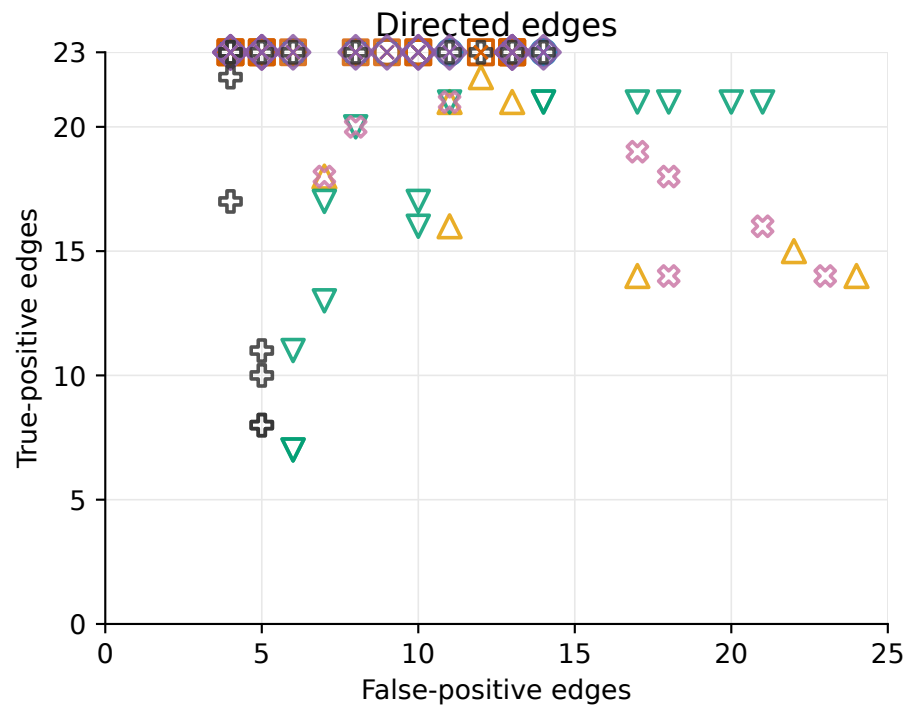
- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

40,000 observations | 10 variables | 23 reference edges | 97/97 successful fits

40,000 observations | 10 variables | 23 reference edges | 97/97 successful fits

0-25 FP viewing window; other points remain in full results



- PS-MIP: complete targets
- ◻ PS-MIP: known-present targets
- ◊ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied
- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

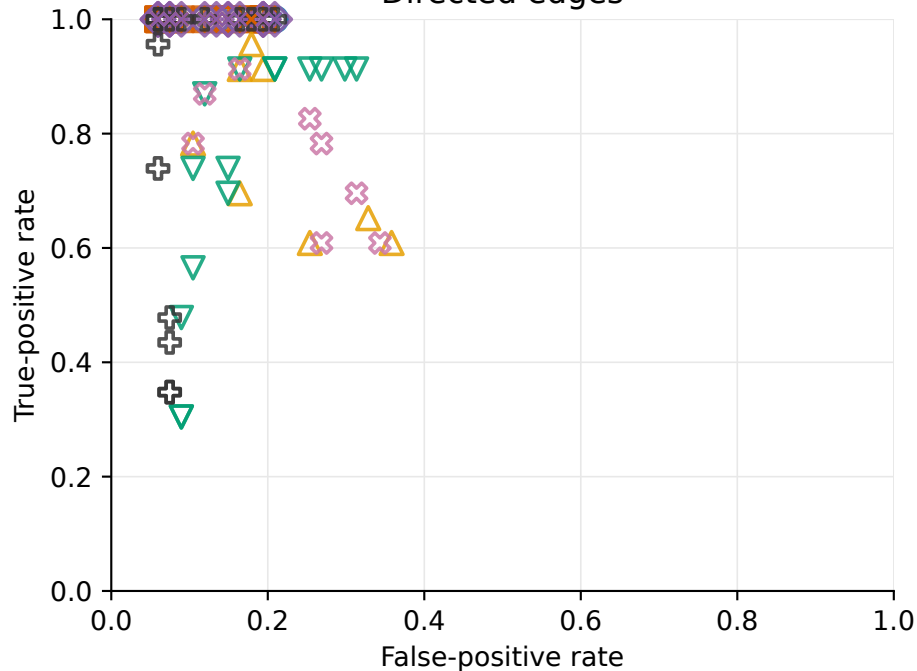
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

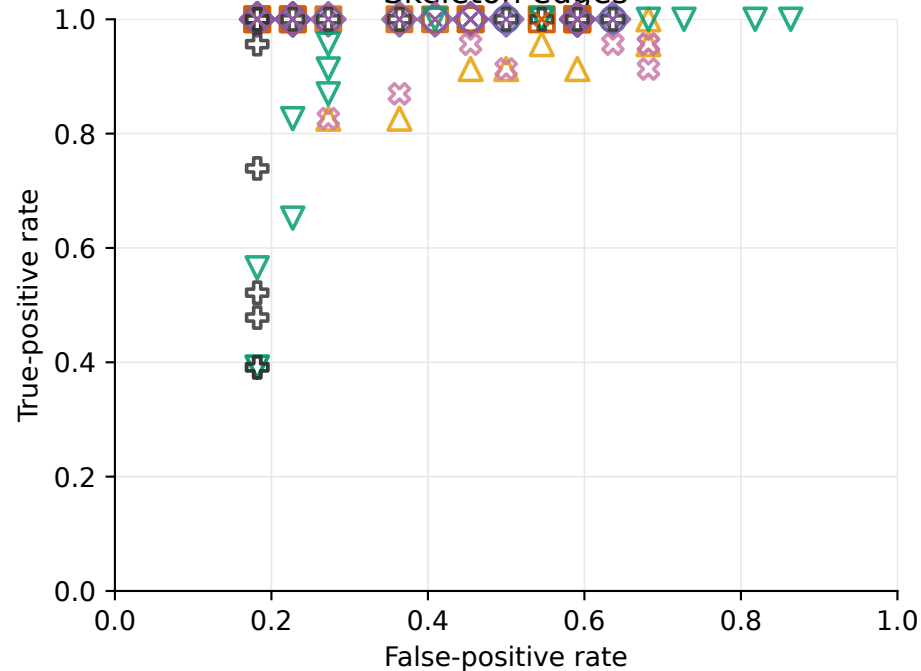
40,000 observations | 10 variables | 23 reference edges | 97/97 successful fits

Full-range normalized paths

Directed edges



Skeleton edges



- PS-MIP: complete targets
- PS-MIP: known-present targets
- ◇ PS-MIP: hidden targets (diagnostic)
- △ UT-IGSP: targets supplied

- ▽ GnIES: target union supplied
- ⊕ GIES: complete targets
- ⊗ IGSP: complete targets

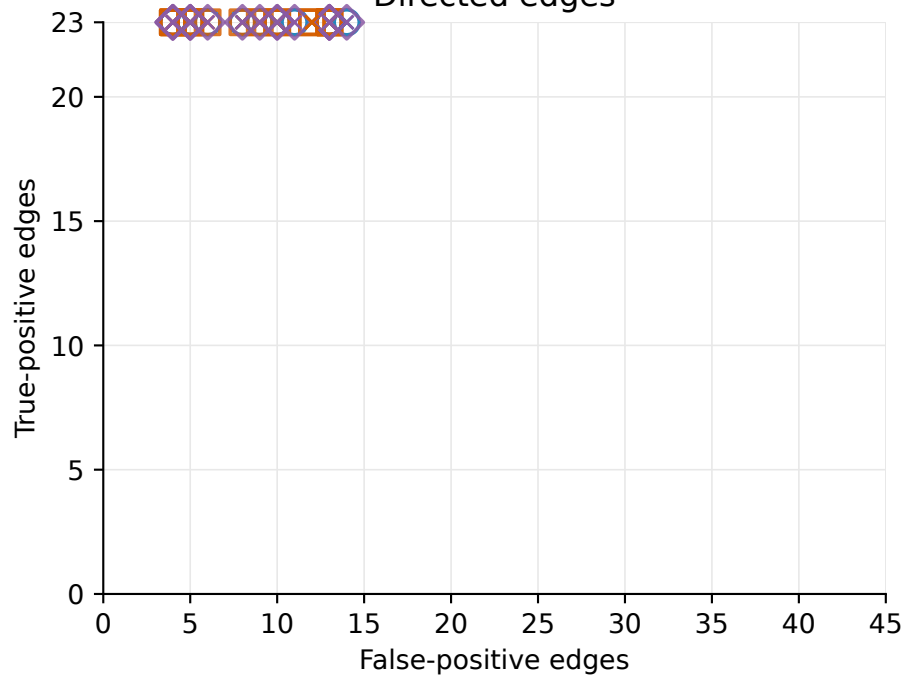
Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

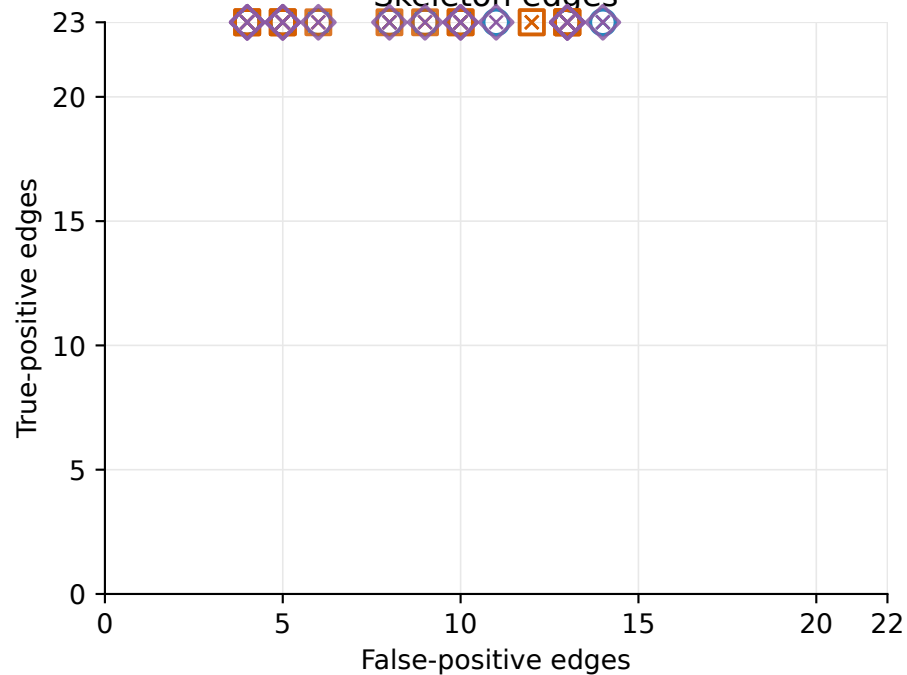
40,000 observations | 10 variables | 23 reference edges | 51/51 successful fits

Full-range paths; every successful operating point retained

Directed edges



Skeleton edges



○ PS-MIP: complete targets

□ PS-MIP: known-present targets

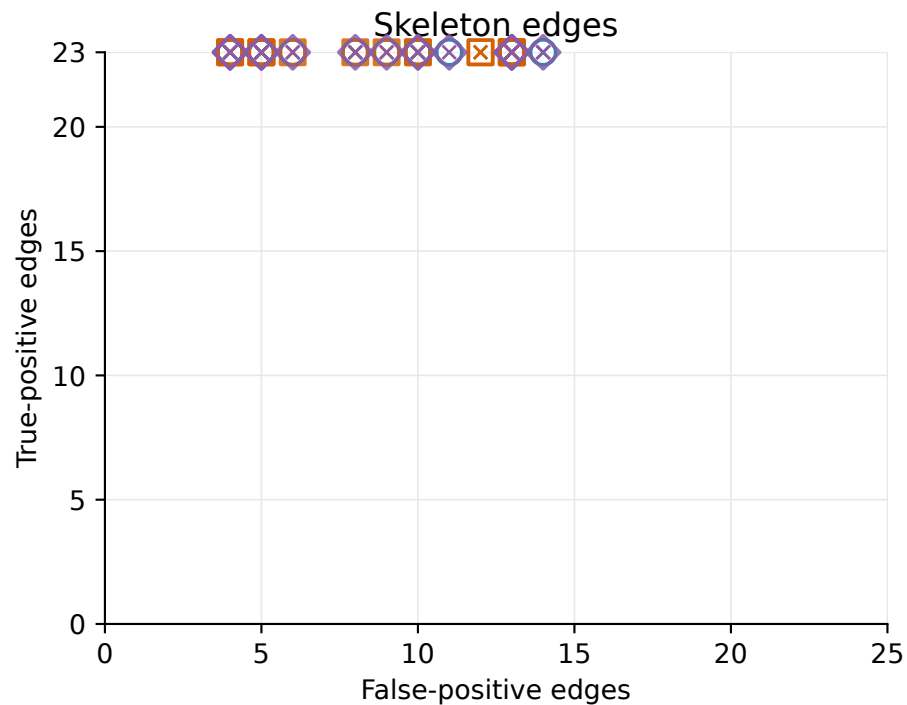
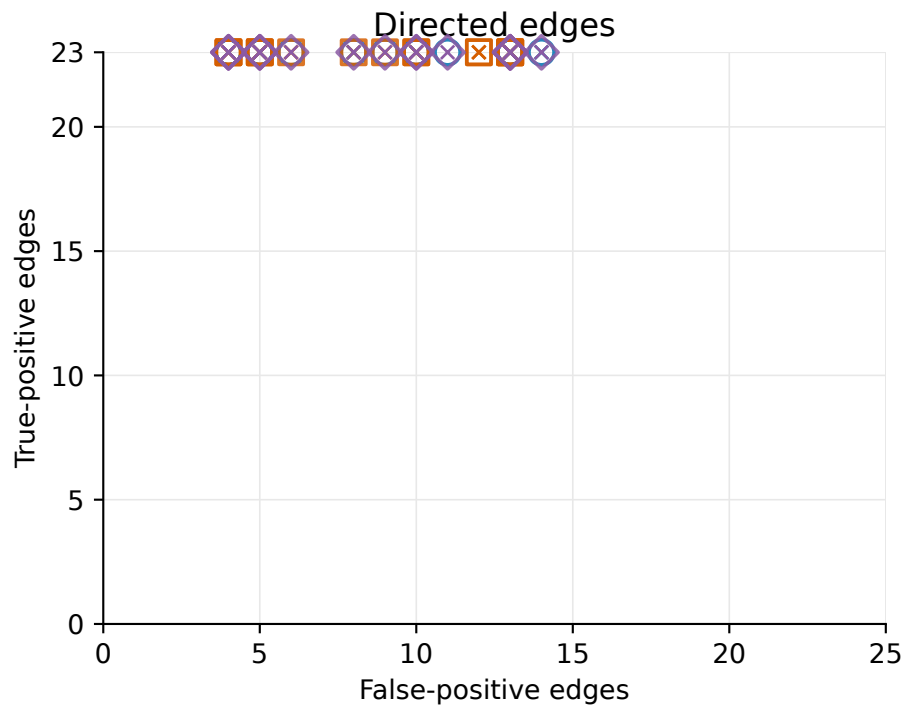
◇ PS-MIP: hidden targets (diagnostic)

Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

40,000 observations | 10 variables | 23 reference edges | 51/51 successful fits

0-25 FP viewing window; other points remain in full results



- PS-MIP: complete targets
- PS-MIP: known-present targets

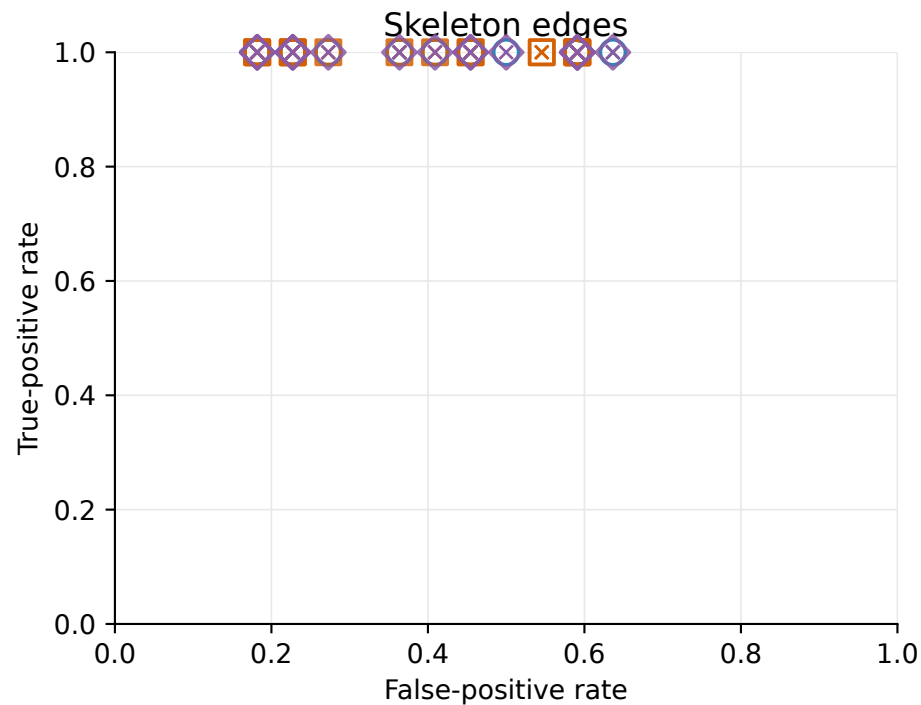
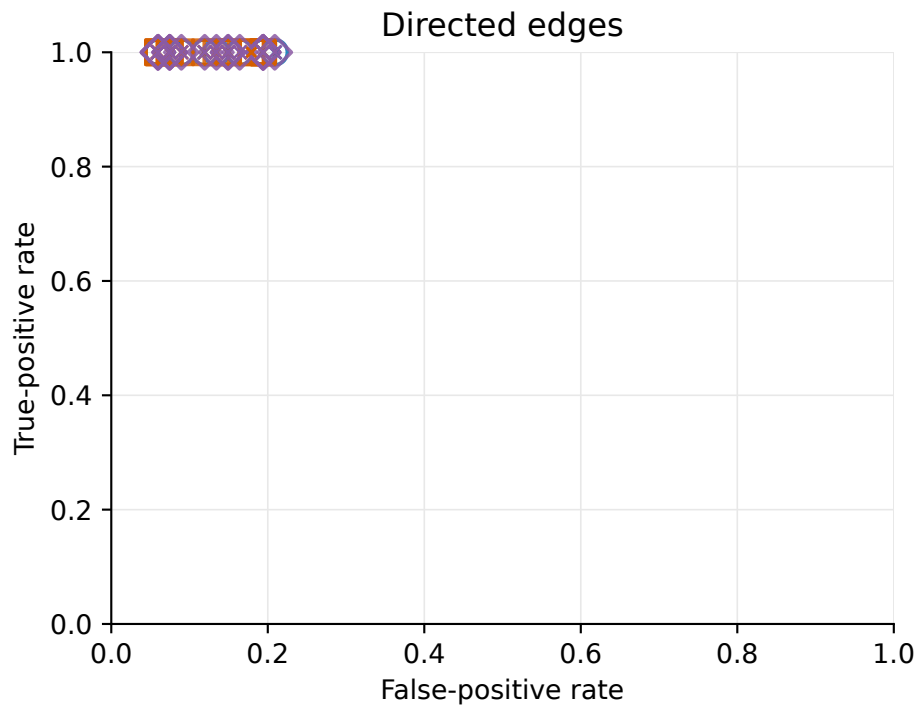
- PS-MIP: hidden targets (diagnostic)

Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Causal Chambers | Actuator chain (scm_4)

40,000 observations | 10 variables | 23 reference edges | 51/51 successful fits

Full-range normalized paths



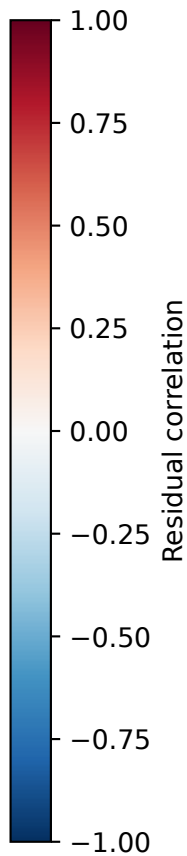
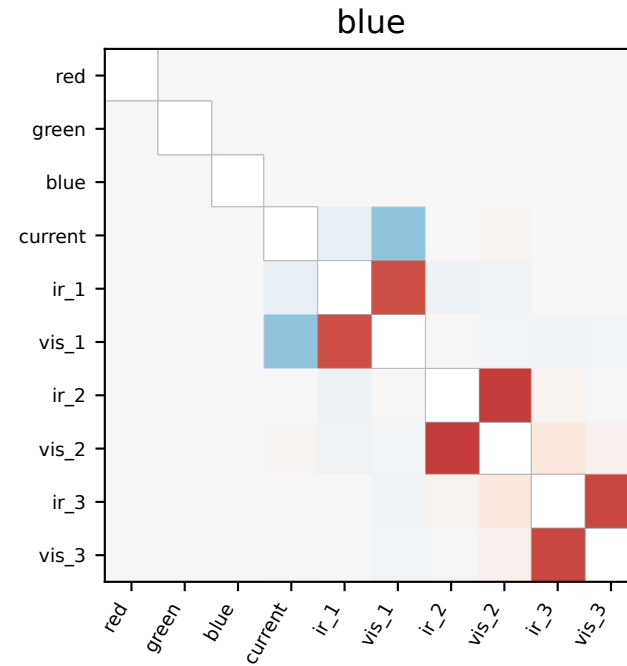
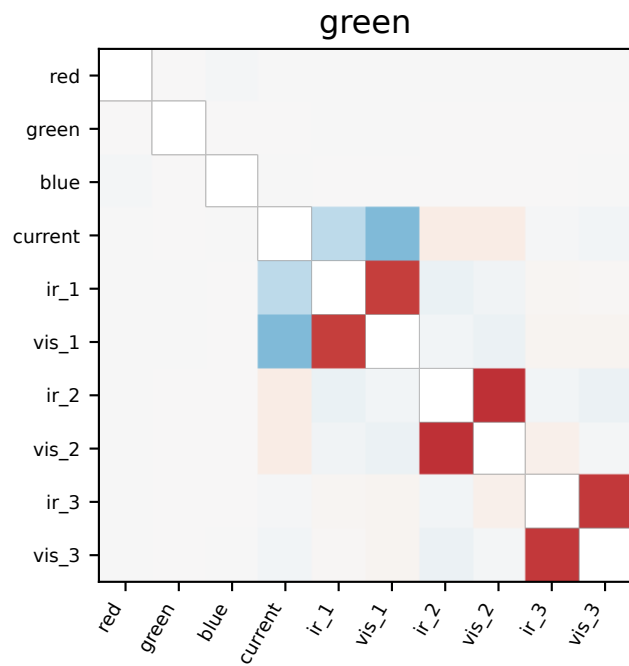
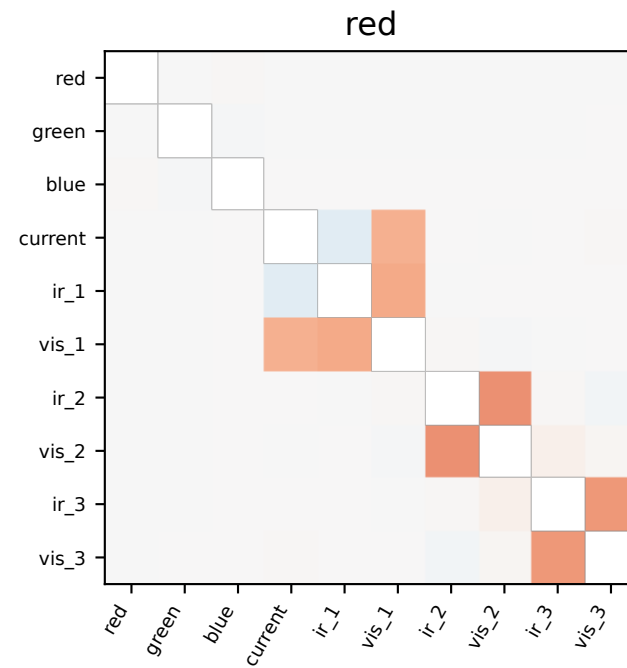
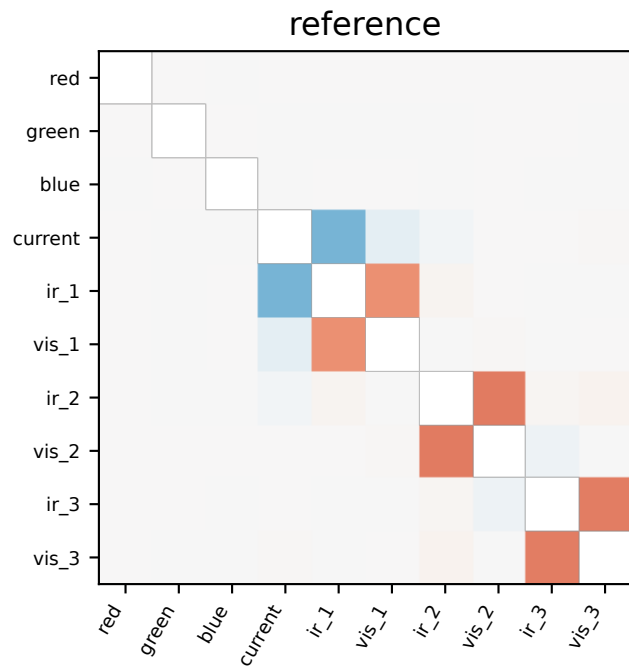
○ PS-MIP: complete targets
□ PS-MIP: known-present targets

◇ PS-MIP: hidden targets (diagnostic)

Unconnected settings; overlaps retained. Cross inside a marker = feasible but nonoptimal PS-MIP.

Model check: scm_4

Residuals after regression on documented parents, fitted separately within each environment



Off-diagonal dependence conflicts with independent SEM errors. No rows or variables were removed.
These diagnostics do not identify a unique cause or establish that omitted physical effects are absent.